



Direct comparison of PMT and SiPM PET systems using modified NEMA IEC Body phantom

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ABSTRACT

Purpose: This study aimed to compare the image quality of the Siemens Biograph mCT40 (photomultiplier-based system – PMT) and the Siemens Vision600 (silicon photomultiplier-based system – SiPM) using a modified NEMA IEC Body phantom (Data Spectrum).

Methods: SiPM-based Vision600 has a smaller crystal size (3.2×3.2 mm vs. 4.0×4.0 mm in the PMT-based mCT40), resulting in better spatial resolution. Enhanced time-of-flight (TOF) timing and higher sensitivity leads to nearly four times higher effective sensitivity. The standard NEMA IEC Body phantom was modified with a 3D-printed holder to accommodate also Hollow and Micro Hollow Spheres of 15.4 mm, 12.4 mm, 7.9 mm, 6.2 mm, 5.0 mm, and 4.0 mm. Each of the three acquisition sessions per scanner included 18 time points and spanned 5.6 half-lives to assess system performance at varying activity concentrations in the field of view.

Results: Recovery curves for both systems were similar when identical post-reconstruction filters were applied. The SiPM-based Vision600 system detected smaller sources at significantly lower activity concentrations, and the variations in standardized uptake value (SUV_{max} , SUV_{A50}) measurements were generally smaller compared to those of the PMT-based system. The two smallest sources became undetectable below 63 MBq and 16 MBq on the PMT system, versus 20 MBq and 6.5 MBq on the SiPM system.

Conclusions: SiPM technology demonstrated superior performance compared to PMT in detecting small sources in low-activity scenarios and provided more robust quantification results. It is recommended to use averaged SUV metrics, such as SUV_{A50} or SUV_{peak} .

1. Introduction

SiPM technology offers promising advancements in PET imaging of small lesions due to its improved spatial resolution and sensitivity [2]. Numerous studies confirm that SiPM technology exceeds the performance of “analog” systems [3–9]. However, these studies typically utilized patient data, the NEMA IEC phantom with a standard set of spheres, or the NEMA IEC phantom with small spheres. Additionally, single realizations of experiments, even with list-mode data, do not allow independent estimation of quantification variations. Single realizations may also introduce systematic errors due to phantom filling inconsistencies. Repeated static measurements ensure no correlation, unlike long list-mode data acquisitions.

The aim of this study was to use a wide range of sphere sizes and multiple acquisitions to simulate activity concentrations in a scanner’s field of view. This approach assessed quantification variations at

different noise levels and independent experiment realizations to minimize inaccuracies related to phantom filling. Results from the Siemens Biograph mCT40 (PMT-based) and the Siemens Biograph Vision600 (SiPM-based) were compared using reconstructions that adhere to EARL v2 criteria [10].

2. Material and methods

2.1. Phantom

A 3D-printed holder was employed to obtain recovery curves for spheres with inner diameters of 37, 28, 22, 17, 15.4, 13, 12.4, 10, 7.9, 6.2, 5, and 4 mm. This holder was fixed around the lung insert of the NEMA IEC Body phantom, aligning the centers of all spheres roughly within the same plane (Fig. 1). Sphere size uncertainties were eliminated using the gravimetric method (RADWAG, AS 22.R2), and the correct

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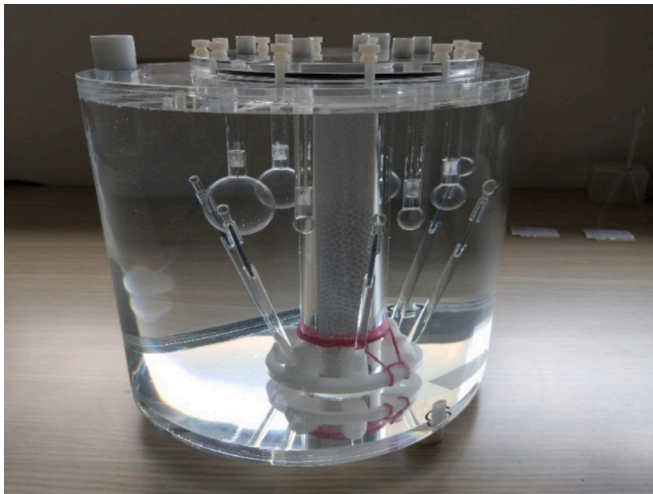


Fig. 1. All 12 sources placed in an IEC Body phantom using 3D printed holder.

inner diameters are listed in Table 1.

The phantom and spheres were filled according to the EARL Image Quality Test procedure [11], targeting a 1:10 background-to-sphere activity ratio. Contrary to the guidelines, the initial activity at the PET acquisition start was set to 100 MBq for the background compartment and 100 MBq in 1000 mL of distilled water for the sources' stock solution (Table 2). Syringes (5 mL each) were prepared using a Karl100 dose administration system (Tema Sinergie, Italy) with a metrologically verified (Czech Metrology Institute) ISOMED2010_T calibrator (model 680000–13). A quarterly cross-calibration between this calibrator and both scanners is performed as part of the EARL accreditation process. Residual ^{18}F -FDG activity in the syringes was eliminated through triple flushing with distilled water during the preparation process.

2.2. Acquisition and reconstruction

Each realization in Table 2 corresponds to a single-bed dynamic mode acquisition. A freshly filled phantom was used in each of the six measurement sessions, with three sessions per scanner. Data acquisition lasted for 6 min, followed by a 30-minute delay before the next time point. This process resulted in 18 time points with decreasing activity concentrations within the field of view.

Throughout the 618-minute duration of each study, there was no movement of either the patient table or the phantom. The phantom was positioned on the couch so that the centers of the spheres were axially aligned with the central slice of the scanner. Additionally, the height was adjusted to ensure that the center of the lung insert was within the transaxial field of view.

The effective sensitivity of the Vision600 ($102 \text{ keps.MBq}^{-1}$)

Table 1

Diameters as declared by the manufacturer and gravimetrically determined sphere diameters.

Declared (mm)	Correct (mm)
37	40.95
28	31.01
22	24.40
17	18.84
15.4	17.22
13	14.24
12.4	13.66
10	10.98
7.9	8.62
6.2	6.91
5	5.46
4	4.28

significantly exceeds that of the mCT40 (keps.MBq^{-1}). Reconstruction parameters meeting EARL v2 criteria are listed in Table 3. For the same acquisition time and volume, differences in pixel volume balanced the noise levels of both systems. This was corroborated by Carlier et al. [6]. To assess potential irregularities observed in low-filtered mCT40 images [1], additional reconstructions of Vision600 data with 2 mm and 3 mm post-reconstruction filters were performed.

2.3. Data analysis

Data analysis was conducted using the EFOMP RC_analyzer tool, developed for the purpose of EFOMP Protocol Quality controls in PET/CT and PET/MRI [12]. This tool is a Python script (Python Software Foundation, <https://www.python.org/>) that employs a region-growing algorithm to delineate the source volume. The user defines a seed by selecting a voxel within the source. The volume delineation procedure is semi-automatic, as the script accommodates any phantom shape and any number of sources without relying on predefined templates for sources or background coordinates. It uses only PET images with all corrections applied (as specified in Table 2) to measure SUV.

The threshold for the region-growing algorithm was set according to the SUV_{A50} definition, i.e., at 50 % of SUV_{max} , while taking the average SUV of the background into account. The background region was automatically defined by a modified version of the original Python script. The average background activity concentration was calculated from the region of interest (ROI) created in the central slice (aligned with the centers of the spheres) and four additional ROIs spanning $\pm$ two slices. The diameter of the background ROIs was 35 voxels for Vision600 and 18 pixels for mCT40 (see Fig. 2). This selection ensured that the transaxial dimensions of the ROIs were approximately the same for both systems, accounting for differences in pixel sizes.

The final recovery curves were generated by averaging the results from three realizations and all time points for each scanner.

3. Results

Both systems demonstrated very similar recovery curves when the same post-reconstruction filtration is applied (Fig. 3).

The recovery curves illustrated that the Vision600 system yielded results closely aligned with those from the mCT40 under EARL v2 reconstructions. However, the true advantage of the SiPM system becomes apparent when narrower post-reconstruction filters are used. Unlike the mCT40 reconstructions, which require sufficient filtration to produce a meaningful recovery curve [1], the Vision600 system allows for less post-reconstruction filtration (e.g., 2 mm). This reduction in filtration enables the Vision600 system to maintain the expected shape of the recovery curve without introducing significant over-corrections.

Consequently, the images obtained from the Vision600 system may provide better contrast, even for small lesions, compared to those from the mCT40 system. If reduced filtration is utilized to enhance lesion visibility in Vision600 reconstructions (e.g., 2 mm instead of 5 mm as per EARL v2), quantitation should be performed using the EQ.PET tool [13] or through dual reconstructions to ensure alignment with the mCT40 recovery curve.

As anticipated, variations in SUV_{max} and SUV_{A50} were larger with narrower post-reconstruction filters (Fig. 3), resulting in noisier images. Nevertheless, the interquartile ranges, defined as the difference between the third quartile and the first quartile, in data from the Vision600 reconstructed according to EARL v2 were generally smaller than those from the mCT40, despite the significantly greater number of image voxels in Vision600.

An F-test was performed to test the null hypothesis that the variance of mCT40 is smaller than or equal to the variance of Vision600 in data reconstructed using EARL v2 compliance parameters (Table 4).

Table 4 shows that, at a significance level of 0.05, the null hypothesis could be rejected for sources with diameters starting from 6.91 mm,

Table 2

Activity values at the time of measurement across all realizations for both scanners. *contained in the background compartment of 9760 ml. **contained in the stock solution of 1000 ml of distilled water.

Biograph mCT40			Biograph Vision600		
Realization	Background activity* [MBq]	Sources activity** [MBq]	Realization	Background activity* [MBq]	Sources activity** [MBq]
#1	99.99	98.16	#1	102.97	101.22
#2	102.03	100.38	#2	101.22	101.76
#3	102.14	100.59	#3	99.17	102.30
average	101.39	99.71	average	101.12	101.76

Table 3

Reconstruction parameters that fulfil the EARL v2 criteria for Vision600 and mCT40.

Reconstruction parameter	Vision600	mCT40
matrix size	440 × 440	256 × 256
pixel size	1.65 × 1.65 mm	3.18 × 3.18 mm
slice thickness	1.65 mm	2.00 mm
# iterations	4	3
# subsets	5	21
reconstruction type	OSEM + PSF + TOF	OSEM + PSF + TOF
post-reconstruction filtration	Gaussian 5 mm	Gaussian 5 mm
applied corrections	attenuation (CT based), randoms, scatter, decay	

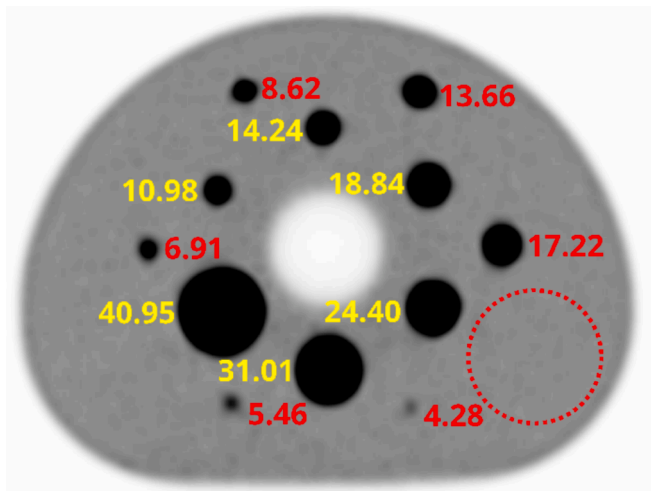


Fig. 2. A representative image from the Vision600 scanner of the reconstructed phantom. The red circle highlights the size and placement of the background ROI. Thresholding was applied to visualize all sources. The diameters in millimeters of the NEMA IEC Body sphere set are displayed in yellow (to the left of the source), while the diameters of selected Hollow and Micro Hollow spheres are shown in red (to the right of the source).

except for the source measuring 10.98 mm. The two smallest sources had in-plane diameters of 2.6 and 3.3 pixels (Vision600) and 1.3 and 1.7 pixels (mCT40). At such small dimensions, whether each pixel is accounted for in the volume plays an important role in the measurement. Despite the fact that, at some activity levels, those sources were visually detectable, measuring any SUV values should be carried out with caution. In a general sense, it can thus be concluded that Vision600 provides measurements with lower variation.

The reason for the significant difference observed in the 10.98 mm source compared to the surrounding sources is unclear. Each realization of the experiment utilized a separate filling procedure, and similar behavior was observed regardless of which scanner was employed. This suggests that the effect was not due to experimental preparation errors. The design of the NEMA IEC Body phantom is intended to minimize any cross-influence between sources. An additional set of sources was positioned to maintain approximately the same distances between their

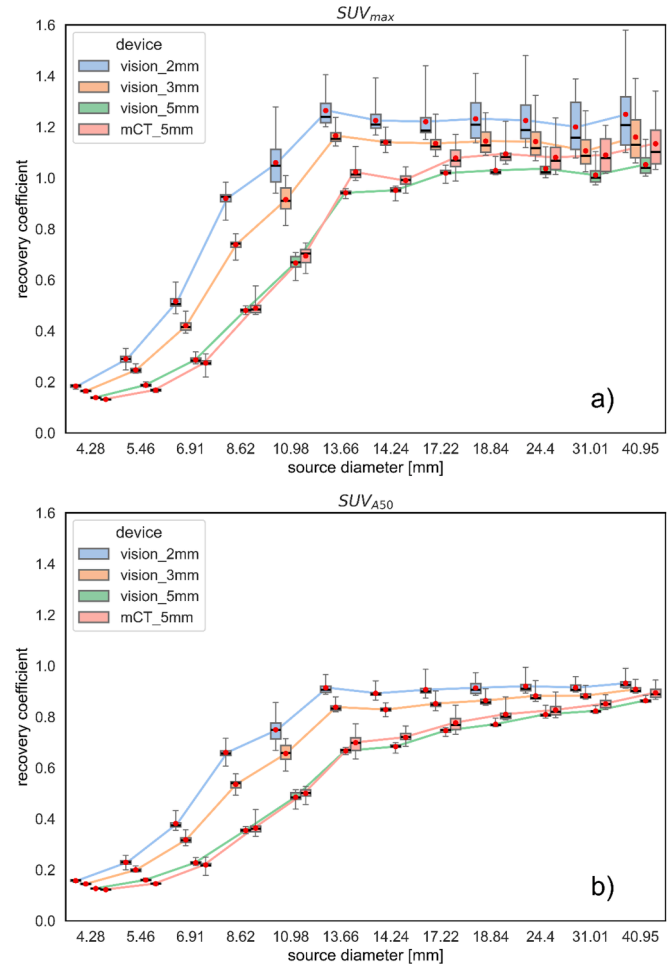


Fig. 3. Recovery curves based on: a) SUV_{max} and b) SUV_{A50} for both systems and various post-reconstruction filters. Each data point in the box plot was constructed using data from three realizations of the experiment and all time points. The box plot displays the mean (red dot), median (black horizontal line), and interquartile range (box), and maximum and minimum values inside 1.5*interquartile range (whiskers). Note that the data points for individual sources are slightly offset for clearer visualization; however, this adjustment may distort the perceived shapes and trends of the recovery curves.

centers and those of the original sources.

While most sources showed an increase in measured SUVs as the activity concentration in the field of view decreases, this specific source exhibited a decrease in its SUV (see Fig. 4). A similar, albeit much weaker, trend was observed in the 14.24 mm source. This phenomenon may be related to the reconstruction algorithm and its ability to handle varying levels of activity concentrations in sources surrounded by other sources. Further exploration of this effect falls beyond the scope of this article, and in non-homogeneous objects (such as patients), this effect would likely be masked by other factors.

Table 4

P-values for the F-test of variances between mCT40 and Vision600 data are provided.

source diameter [mm]	SUV _{max} – 5 mm	SUV _{A50} – 5 mm
4,28	4,92E-01	2,44E-01
5,46	2,87E-01	1,92E-01
6,91	1,25E-02	3,73E-03
8,62	1,58E-05	5,69E-06
10,98	3,05E-01	2,73E-01
13,66	3,15E-06	1,27E-07
14,24	1,45E-03	1,56E-03
17,22	2,57E-05	2,66E-07
18,84	1,04E-03	1,86E-04
24,40	1,40E-02	1,60E-03
31,01	5,71E-03	3,34E-04
40,95	7,76E-04	2,29E-05

Considering an injected activity of 230 MBq of ^{18}F -FDG for a 70 kg patient, it can be estimated that, after a one-hour uptake period with 10 % excretion via urine [14], about 140 MBq remains at the time of examination. Assuming homogeneous distribution and considering that the human torso represents roughly 55 % of body mass, 70 MBq would be present in 39 kg of the patient's body. The volume of the NEMA IEC Body phantom is approximately 10 L.

The acquisition in this study lasted 6 min per bed position, compared to the usual 3 min for a 70 kg patient. Therefore, an activity of 9 MBq in the phantom represented a background volume activity comparable to that observed during a clinical exam. Naturally, this value is lower for radionuclides with lower initial activities or faster decay rates, such as ^{68}Ga .

Fig. 4 also illustrates that lowering the activity concentration within the field of view increased the SUV_{max} measurements from both scanners. This effect was particularly prominent in large sources because they contain a greater number of voxels. Fluctuations in recovery

coefficients were smaller for the SiPM system due to decreased image noise. For cases of low activity in the field of view, measurement techniques such as SUV_{A50} or SUV_{peak} should be considered for both scanners.

The smallest sources could not be detected at low activities (Fig. 5) because of insufficient source-to-background volume activity contrast, which occurs much earlier in the PMT system.

4. Conclusion

Measurements with 12 active sources in the NEMA IEC Body phantom, covering diameters from 41 mm down to 4 mm, were conducted for both the SiPM and PMT systems. Different noise levels in the resulting images were achieved through the physical decay of ^{18}F over more than five half-lives.

The Vision600 system is capable of maintaining its excellent resolution by reconstructing images into a 440×440 matrix while simultaneously reducing the amount of injected activity. Images from the Vision600 system exhibited fewer fluctuations in recovery coefficients, leading to more reliable results compared to the mCT40 system. This is attributed to the significantly improved TOF resolution used in the reconstruction process. However, the increase in the number of voxels in the reconstruction to preserve the spatial resolution should be taken into considered. A simple reduction in injected activity, based solely on the difference in NEMA sensitivity or effective sensitivity, without other modifications to acquisition and reconstruction protocols could easily result in inferior images.

The results indicate that, even when both systems meet EARL v2 accreditation criteria, the SiPM system outperforms the PMT system in detecting small sources in low-activity situations and in maintaining stable quantification results.

The data further suggest that in low-activity acquisitions, the use of SUV “averaging” metrics (e.g., SUV_{A50} (SUV_{peak})) is more reliable. For

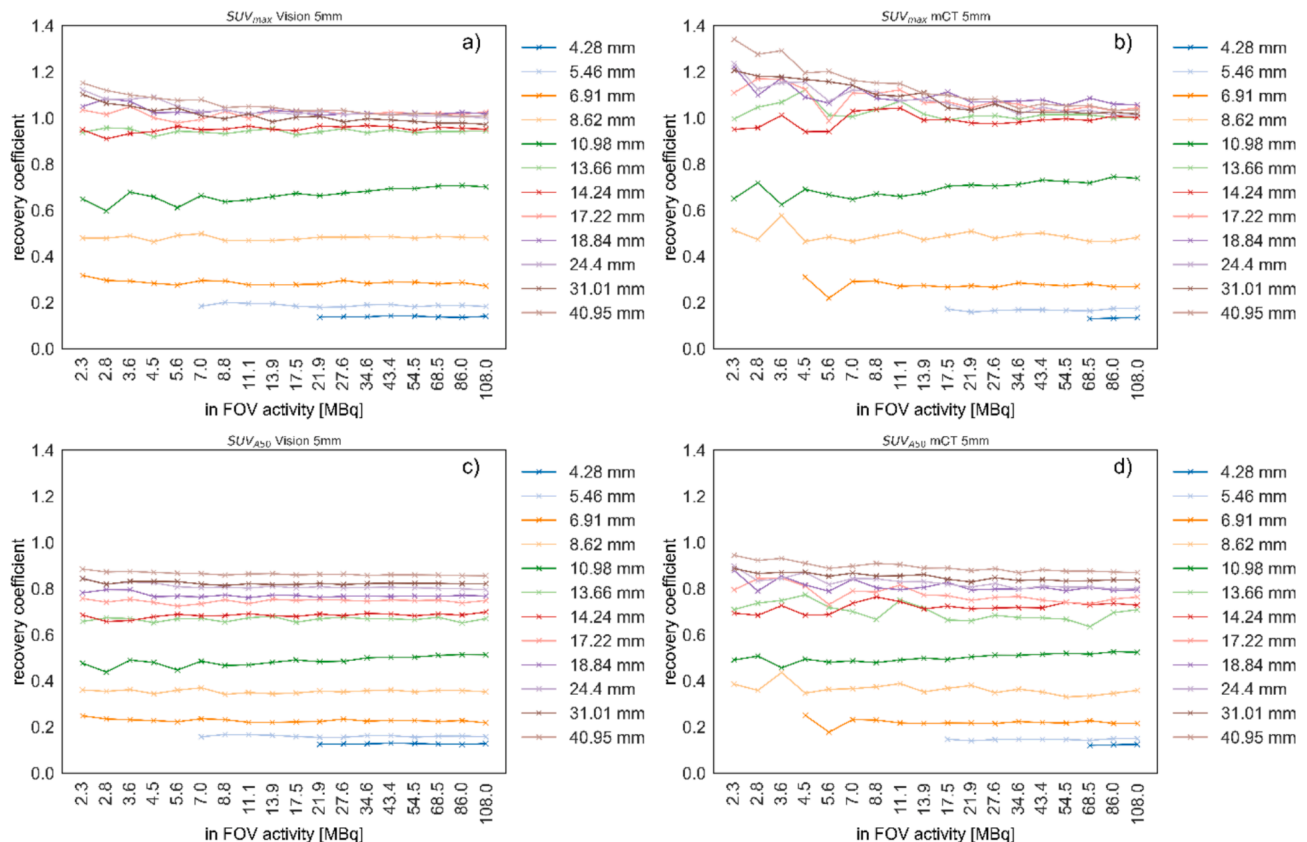


Fig. 4. Recovery coefficients at varying activity concentrations within the field of view.

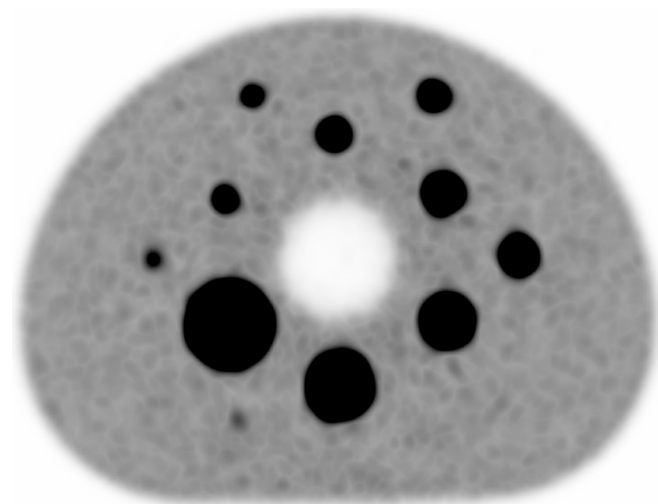


Fig. 5. Insufficient source-to-background contrast prevented the smallest source from being detected. The EFOMP RC analyzer tool requires the operator to select the seed voxel for the region-growing algorithm rather than using a predefined mask of source positions. If a source remains undetectable even after applying image thresholding, delineation cannot be performed.

larger sources, the higher number of pixels increases the likelihood of SUV_{max} values being influenced by image noise. Using SUV “averaging” improves the reliability of SUV measurements when comparing patient studies acquired under different conditions, such as varying injected activity or bed duration.

This study has limitations due to the use of only a single sphere-to-background ratio (SBR) of 10:1. Smaller SBRs may result in smaller sources becoming undetectable at an earlier stage. Nevertheless, the conclusion that the SiPM system was superior to the PMT system remains valid. The considerably higher P-values for the F-test of variances between mCT40 and Vision600 data for the 10.98 mm source were not clarified and would require more thorough investigation. Any trends in data subjected to the F-test may influence the results, as normally distributed input data is required.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

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